RailNetwork game

- user manual -

designed and developed by Codrin Berinde
student, Technical University of Cluj-Napoca, Romania
semestrial assignment for Computer Architectures course (B35APO)
at the Faculty of Electrical Engineering
Czech Technical University in Prague
coordinating professor: Pavel Píša
31st May 2025

1 Game description

The project is a game of 1 player, so that anyone can play it when feeling lonely.

1.1 The board

The game features a rectangular board of 12×18 cells. Each cell can either be empty, or a number can be found written in it. The number represents the population of a city that lays in the respective cell. Any city may have a population of minimum 1 to maximum 15. A sample board is shown in Figure 1.

							4										
				1													
												3					
		2															
				3													
													9				

Figure 1: Sample board

The player also has a certain number of initial points. The player has to build a rail network that connects all the cities on the board. A railway is built segment by segment. Each segment starts in the middle of one cell and ends in the middle of an adjacent cell (an adjacent cell is a cell that shares an entire edge with it; no diagonal rail segments may be built). Each segment that is built costs the player 1 point down. Once placed, a segment cannot be removed. Figure 2 shows some built segments on the board.

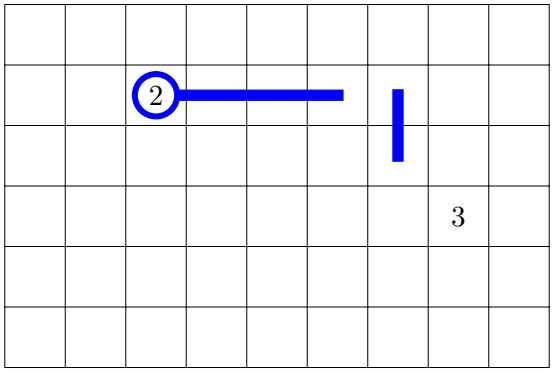


Figure 2: Sample segments

In the figure, 3 contiguous segments were built starting with city with population 2. As the reader may notice, when a segment is connected to a city, it is drawn as a ring around the number, but this is only a matter of representation, the functionality is just the same as for the other segments. Then another vertical segment was built, but it is not connected to the others.

1.2 Gaining points

Once a contiguous set of segments (a railway) was built between two cities, the player gains points equal to the sum of the populations of the two cities. However, a city may provide points only once in a game: the first time when it was connected to some other city/cities. For example, Figure 3 shows in dark blue the connection of the cities with population 2 and 3, respectively. Upon this connection, the player gains $2 + 3 = 5$ points. Later, the connection with the city 4 is shown in lighter blue. Upon that connection, the player will only gain 4 points because 4 is the citythat was connected for the first time, but the others (2 and 3) had been previously connected.

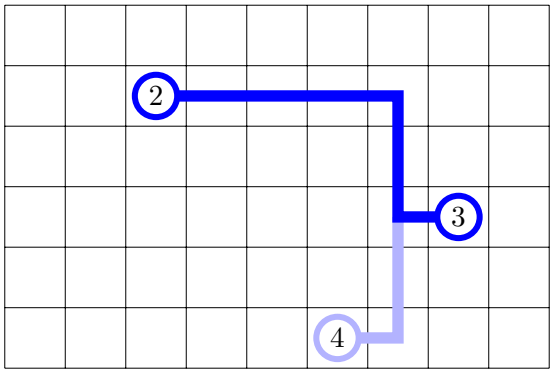


Figure 3: Gaining points mechanics (1)

Another example is shown in Figure 4, where let's say that the player first connected cities 1 and 2 and gained $1 + 2 = 3$ points, then connected cities 3 and 4 and gained $3 + 4 = 7$ points, but at the end they connected the two subnetworks with each other, but gained 0

points because there was no city to be connected for the first time to another.

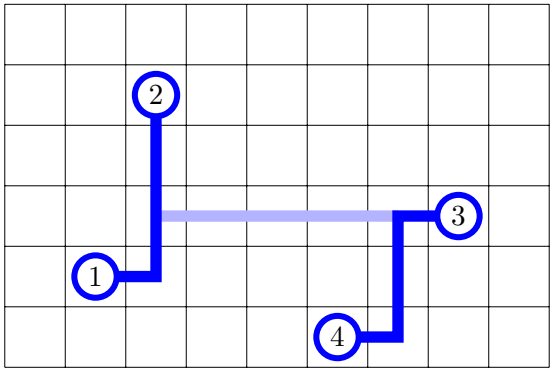


Figure 4: Gaining points mechanics (2)

1.3 Game end

The game is won when a single network connecting all cities exists on the board (i.e. when between any two cities on a map there is a contiguous set of rail segments connecting them - even if it passes through other cities) and is lost when the player remains with 0 points but the above condition is not met.

2 Gameplay

The game is played on the Educational Kit MicroZed APO, that can be observed in Figure 5.

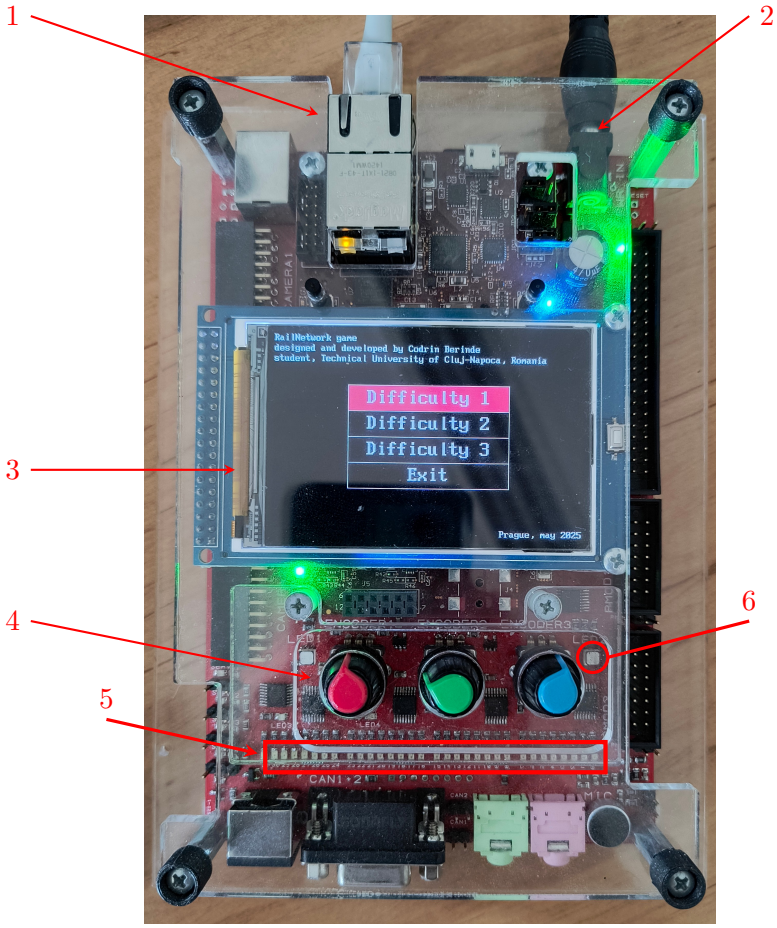


Figure 5: The MicroZed APO Educational Board

The game uses the following features of the board

1. Ethernet port
2. Supply jack
3. LCD screen
4. a red, a green and a blue knob that can be pushed and rotated
5. a strip with 32 yellow LEDs
6. a RGB led

2.1 Connection to the board

The program can be run on the board via ethernet. Plug the ethernet cable into the board and wait a few seconds until the board displays its own IP address. The IP address is displayed as 4 rows of 8 rectangles each, every rectangle can be black - corresponds to binary value 1, or white - value 0. convert the IP address into decimal (the first 3 bytes will be 192.168.223) and run the following commands from the computer which is connected in the same network

```
ssh-add-mzapo-key
ssh root@192.168.223.XXX
```

where `XXX` should be replaced with the value read on the board.

Then, go to the location of the `Makefile` of this project on the computer and run

```
make TARGET_IP=192.168.223.XXX run
```

where again `XXX` should be replaced with the actual value.

On the computer terminal, the moment when you see "Game started", it means that the game runs correctly.

2.2 Gameplay

When the game is run, the main menu appears on the LCD screen. The user can use either the red or the green knob to scroll up and down on the menu. The first three options: *Difficulty 1*, *Difficulty 2* and *Difficulty 3* will initiate a new game with the selected level of difficulty. To select an option, the user may press any of the red or green knob. Obviously, option *Exit* will terminate the whole process. By pressing the blue knob the process is terminated, so don't press it!

Once a game is initiated, the board will appear on the LCD screen, the current number of points that the player has will be shown on the LED strip: the number of LEDs turned on equals the number of points. Also, the RGB led informs the player about the status of their points - the greener the led, the more points they have and the more red the led is, the fewer the points.

On the board there is always a cell that is coloured in red - see example in Figure 6. That cell is the current cell in which the player is situated. The player may move to other cells: up and down - by rotating the red knob, and left and right - by rotating the green knob.

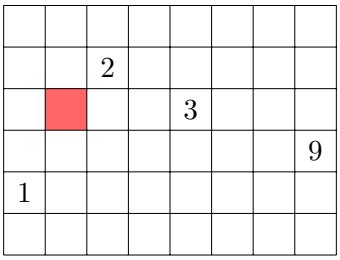
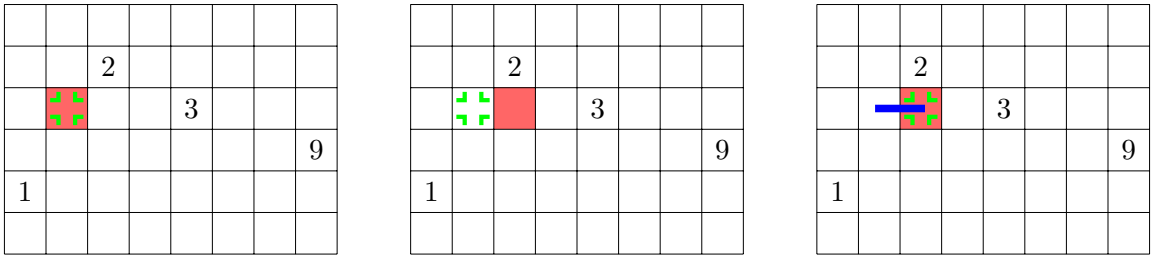


Figure 6: Current cell example



a) b) c)

Figure 7: Building a segment

Once the player decides to build a rail segment, they press on either the red or the green knob and that cell turns into *under construction* cell. This is marked by a green "target" - see Figure 7 a), and it represents the starting cell of the segment the player will build.

After the *under construction* cell is selected, the player moves to another current cell - the end of the segment the player wants to build - see Figure 7 b).

If the *under construction* cell and the current cell are adjacent and the player presses either the green or the red knob, then a segment will be built between the two. Simultaneously, the *under construction* cell is moved to the end of the newly-built segment - see Figure 7 c). It was made like this in order to ease the construction of a longer railway and not having to press on each cell two times.

If however the *under construction* cell and the current cell are not adjacent, but one of the red or green knobs was pressed, than no segment will be built, just the *under construction* cell will be moved to the new location (the current cell).

If the current cell and the *under construction* cell coincide and one of the two knobs is pressed again, the *under construction* cell is deselected and there will be no current cell under construction.

Watch out! The number of points decreases by 1 for each built segment. After each step, this is updated on the LED strip.

By pressing the blue knob during the game, the game is aborted and the player will be redirected to the main menu.

Have fun and good luck connecting all the cities!